# Overcoming Limited Fault Data: Intermittent Fault Detection in Analog Circuits via Improved GAN

Xiaoyu Fang, Jianfeng Qu, Bowen Liu, and Yi Chai

***Abstract*— Owing to the characteristics of intermittent faults (IFs), such as short duration and randomness in analog circuits, it is difficult to collect a sufficient amount of manually labeled fault data for training detection model. To this end, this article explores the application of generative adversarial network (GAN) in the field of IF detection for analog circuits. The proposed model consists of two networks, an autoencoder and a discriminator, which are trained by competing against each other while working together to grasp the underlying concepts in the target. In addition, a spatial Fourier convolution (SFC) block is proposed and introduced into the discriminator to enhance the detection performance of the model. The training process is divided into two stages, the first stage through the autoencoder and the discriminator of the adversarial learning, so that the discriminator has the initial fault detection ability. The second stage trains the autoencoder and the discriminator with a few fault samples to enhance the fault detection capability of the two networks working cooperatively. The proposed method is applied to two typical analog circuits, namely, a power amplifier circuit and a low-pass filter circuit. The results show that, guided by only a small number of fault samples, the proposed method has a great advantage over other state-of-the-art detection methods.**

***Index Terms*— Analog circuits, generative adversarial network (GAN), intermittent fault (IF) detection, limited fault data, spatial Fourier convolution (SFC) block.**

## I. INTRODUCTION

ANALOG circuits, being essential components of complex electrical instruments, perform the main tasks of signal generation, arithmetic operations, signal conversion, power amplification, and signal filtering [1], [2], [3], [4], [5]. In electronic systems with mixed analog–digital circuits, the analog circuits comprise less than 20% of the total, yet they are responsible for approximately 80% of faults [5], [6], [7], [8]. Moreover, intermittent faults (IFs) in circuits tend to occur approximately 10–30 times more frequently than permanent faults (PFs) [9]. Failure to detect IFs early can result in severe harm to the system when they progress into PFs [10], [11]. Thus, precise measurement of IFs in analog circuits within electronic systems is of utmost importance.

An IF is a fault that reoccurs at the same location within a short duration and is caused by the same reason [12]. IFs are primarily caused by temporary interruptions in the circuit. Examples of such interruptions include loose solder joints, corroded or oxidized connections, and exposure to extreme environments [13], [14]. In the solder joint of a circuit, poor contact leads to the formation of electrical contact resistance (ECR) at the component connection [15]. The ECR intermittently manifests under the influence of vibration. IFs are cumulative, as ECRs gradually build up over time, leading to eventual complete breaks and circuit shutdowns [16], [17]. Hubner-Obenland and Minuth [18] discovered that the majority of ECRs are more than two orders of magnitude higher, with durations mostly in milliseconds. According to the U.S. military's standard definition, millisecond IFs for electronic systems have a time duration not exceeding 5 ms, and the associated ECR is higher than 500 Ω [19].

In general, IF diagnosis comprises two aspects: IF detection and IF recognition [15], [20]. Due to the short duration of IFs, the acquired signals contain numerous redundant fault-free signals. Therefore, the initial step involves detecting the moments of appearance and disappearance of IFs using fault detection methods and subsequently extracting the IF signals for fault recognition purposes. To detect the appearance and disappearance of IFs, the length of the detection window is typically required to be smaller than the fault signal length [21]. Some scholars have conducted research related to IF detection. Huang et al. [22] proposed a fractional-order transient chaotic system for intermittent sensor fault detection. Sheng et al. [23] used the reduced-order observer to detect IFs for linear discrete-time stochastic multiagent systems. Zhang et al. [24] used the lifting method and a reduced-order observer for the IF detection of discrete-time linear stochastic systems with time delay. Nonetheless, analytical models often fail to fully represent the complexity of the actual system and are challenging to accurately model.

Data-driven approaches for IF detection, which do not require the construction of precise mathematical models, are gaining the interest of researchers. Zhong et al. [25] used the Fourier transform to detect IFs in analog circuits. Shen et al. [26] proposed a wavelet packet transform (WPT)-based denoising method for IF detection. Melani et al.'s [27] principal component analysis (PCA) was used to IF detection. Zhao et al. [28] proposed a moving average (MA) method to detect IFs. Tian et al. [29] proposed a $k$-nearest neigh-

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bor (KNN) distance for IF detection. Nonetheless, shallow machine learning models typically rely on manually designed signal processing and feature extraction methods, which may lack generalizability. Moreover, it becomes challenging to identify an effective feature extraction scheme when there is insufficient knowledge of the system's fault mechanism.

End-to-end deep learning methods are gaining attention in the detection field. Moreover, researchers have made efforts due to the limited accessibility of fault anomaly data. González-Muñiz [30] proposed a deep variational autoencoder (DVAE) for fault detection. Fang et al. [31] proposed an adaptive multiscale and dual subnet convolutional autoencoder (AMDSCAE) to detect IFs of analog circuits in noise environment. These autoencoder models solely rely on fault-free data for training. As a result, when faulty data are encountered during detection, the models generate significant reconstruction errors, which can be exploited for fault detection. However, the high capacity of deep learning also results in improved reconstruction of certain fault signals, which, in turn, impacts the detection performance. To solve this problem, a prior knowledge-guided teacher–student (PKGTS) model based on self-supervised learning was proposed for IF detection of analog circuits [32]. By introducing prior knowledge, the detection residuals of fault signals are increased, and the problem of limited fault data is alleviated. Similarly, Wang et al. [33] employed a self-supervised signal representation learning (SSRL) approach to improve the model's comprehension of the signal. In addition, they fine-tuned the model with limited fault samples. Nonetheless, both the PKGTS and SSRL models still rely on human intervention to design the prior signal transformation scheme.

Recently, generative adversarial networks (GANs) have offered fresh insights in the field of anomaly and fault detection, addressing the challenge of detecting faults with limited data. The GAN's discriminator attains its initial detection capability by distinguishing between real data and fake data generated by the generator, and this process solely relies on fault-free data. Different from DVAE [30] and AMDSCAE [31], GAN not only utilizes the generator reconstruction bias to detect faults, but also utilizes the detection capability of the discriminator, and the two work in concert to further enhance the detection capability. Sabokrou et al. [34] proposed adversarially learned one-class classifier (ALOCC) for anomaly detection. Jiang et al. [35] proposed a GAN-based anomaly detection (GBAD) approach to solve the data imbalance problem with limited anomalous data. These two GAN-based detection methods collaborate for detection through generators and discriminators. However, the generator goal in the detection problem is to make the fake samples as close as possible to the real normal samples, which may cause the discriminator to confuse the normal samples with the fake samples, thus resulting in poor detection. Zaigham Zaheer et al. [36] proposed a old is gold (OIG) framework, which leverages low-quality data generated by the generator during the early training epochs as pseudo-anomaly samples to enhance the detection capability of the discriminator. However, this method is mainly aimed at image anomaly detection, and the single-scale local convolution used makes it difficult to rapidly increase the receptive field. Therefore, it is necessary to develop an effective GAN-based detection method for analog circuits.

A well-designed architecture should incorporate units with the broadest possible receptive field at an early stage. For example, it is believed in [31] that large-scale convolutions are advantageous for noise suppression, while small-scale convolutions are useful for capturing local details. Consequently, a three-branch multiscale convolution fusion module is proposed to enhance the IF detection capability in analog circuits. Moreover, Ding et al. [37] demonstrated that using oversized kernel convolution is advantageous in enhancing model accuracy. However, given the short duration of IFs, the detection algorithm needs to achieve high efficiency, and large kernel convolutions might not be suitable for computationally efficient scenarios. Therefore, in this article, a spatial Fourier convolution (SFC) block is proposed and introduced into the GAN's discriminator. Different from [31], the computational burden of acquiring large receptive fields can be reduced by the fast algorithm, fast Fourier transform (FFT). Shi et al. [38] and Kumawat et al. [39] employ Fourier transform block, but the proposed SFC block still exhibits differences. Different from [38], which obtains global information solely through the FFT block, the proposed SFC block not only captures global information but also supplements local information, potentially resulting in a richer set of information. In [39], it is considered that low-frequency components contain most of the information, and thus, a short-term Fourier transform (STFT) block is used to combine low-frequency components to obtain a new information representation. However, this may not be applicable to analog circuits, e.g., the signals in some high-pass filtering circuits are usually high frequency. Therefore, unlike [39], our designed SFC block filters real and imaginary parts through maximum pooling. We believe that information with larger amplitudes is valuable, because noise amplitudes in circuits are usually smaller than signal amplitudes. Furthermore, [39] is primarily designed for action recognition and cannot be directly applied to the field of analog circuits. Developing targeted IF detection methods for analog circuits is of significant importance.

In this article, a new detection framework is designed based on the characteristics of IFs in analog circuits. The proposed framework contains pretraining of GAN and fine-tuning with a small amount of fault data. The proposed training strategy enables enhanced collaboration between the generator and the discriminator, resulting in improved IF detection capabilities. Furthermore, the SFC block enables the model to achieve coverage of the signal's receptive field at relatively shallow layers through the use of the FFT. Moreover, SFC operates in both the spatial and Fourier domains, thereby enhancing the model's noise robustness and feature capturing ability.

The contributions of this article are summarized as follows.

1) To tackle the issue of limited IF data in analog circuits, a novel detection framework is proposed that utilizes a small amount of fault data-guided GAN.

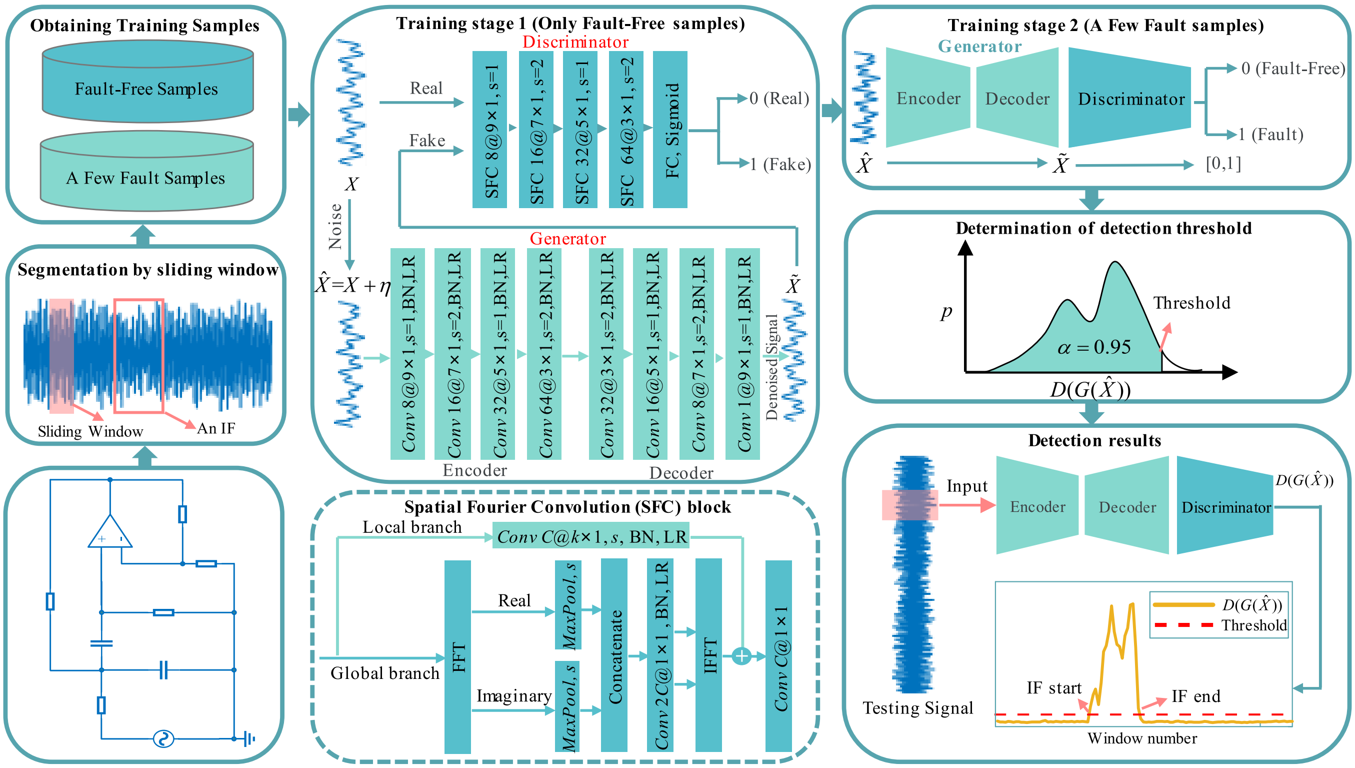


Fig. 1. Proposed GAN-based detection framework. The "conv $C@k \times j, s$" represents that the 1-D convolution kernel size is "$k \times j$," the number of channels is $C$, and the stride of the convolution is $s$. LR represents activation function LeakyReLU. BN represents batch normalization.

2) The proposed two-stage training strategy effectively facilitates the collaboration between the generator and the discriminator during the testing phase, enhancing the model's sensitivity to fault signals.
3) The proposed SFC block learns the local details and global information of the signal through the spatial and Fourier domains, respectively, which contributes to enhancing the detection performance and noise robustness.

The rest of this article is organized as follows. Section II presents the details of the proposed IF detection framework. Section III validates the proposed framework. Section IV provides a summary of the entire article.

## II. Methodology

### A. Overview GAN-Based Detection Framework

Due to limited data on IFs and tendency to be masked by noise, a two-stage training strategy is proposed to address the issue of limited fault data, while the SFC block enhances the model's detection performance and noise robustness. GANs are built upon a two-player game involving two distinct networks, which are trained simultaneously in an unsupervised manner. One of the networks is called generator, which consists of an encoder and a decoder to produce realistic data, such as circuit signals. The other network functions as the discriminator and aims to differentiate between the real data and the fake generated data. Fig. 1 illustrates the detailed structure of the proposed framework.

First, the voltage signals from the circuit measurement point are gathered, and then, the signals are segmented using a sliding window. As outlined in [21] and [23], it is recommended that the window length be shorter than the IF duration. Therefore, the faulty sample should include part of fault signal. In the context of this article's investigation into millisecond IFs, the sliding window duration must be under 1 ms. The segmented signal contains a large number of fault-free samples and a small number of faulty samples. Next, during the first stage of training, only fault-free samples are utilized. The generator functions as a denoising autoencoder to improve noise robustness, and its output is treated as fake by the discriminator. On the other hand, the discriminator treats clean fault-free samples as real sample. This process provides the GAN with an initial fault detection capability, demonstrated by the fact that when a faulty sample is fed into the generator, since the generator has not been exposed to faulty samples during the initial training, the generated sample differs from the real fault-free samples, leading the discriminator to identify it as a fake, indicating a fault. During the second training stage, the generator and discriminator collaborate for the fault detection task instead of undergoing adversarial training. In this stage, only a small number of fault-free and faulty samples are utilized to guide the training of the GAN. The generator only ensures accurate reconstruction of fault-free samples, while the discriminator strives to differentiate between the reconstruction results of fault-free and faulty samples generated by the generator. This aims to strengthen the discriminator's ability to recognize faults by further refining the generator's bad reconstruction of fault samples. Finally, a threshold is determined based on the likelihood of faults in the discriminator output. Finally, detection can be accomplished by threshold comparison during the testing stage.

### B. Network Architecture

*1) GAN:* The detailed parameters of GAN are given in Fig. 1. The generator is a denoising autoencoder consisting of 1-D convolution, batch normalization (BN), and activation function LeakyReLU. The 1-D convolution extracts features along the time direction for voltage signals. The discriminator is stacked by SFC blocks to enhance the ability to identify faults. Finally, the discriminator output, i.e., the likelihood that the current sample is faulty, is obtained from the fully connected layer and the sigmoid activation function.

*2) SFC Block:* The SFC block is based on FFT and 1-D convolution and can obtain information about the receptive field covering the whole signal and local details. SFC divides the channel into two parallel branches: a global branch and a local branch. The local branch employs conventional convolution, while the global branch utilizes FFT to consider the global context. In addition, noise signals and circuit signals often have different frequency bands, and the FFT transform can obtain the information of different frequency bands, which is conducive to improving the noise robustness. Assume that the signal with noise is denoted as $\hat{X} \in \mathbb{R}^{C\times L}$, where $C$ denotes the number of channels and $L$ denotes the length. The calculation process of SFC can be expressed as follows.

In the global branch of the Fourier domain, the real part $\hat{X}_{\text{real}} \in \mathbb{R}^{C\times(L/2)}$ and imaginary part $\hat{X}_{\text{img}} \in \mathbb{R}^{C\times(L/2)}$ of the spectrum are obtained by the FFT. Because the real and imaginary parts exhibit redundant symmetry, we extract the first half sequences of the real and imaginary parts as inputs to the subsequent modules, aiming to reduce the computational complexity of the subsequent computations.

Since noise increases the complexity of the analysis, Fourier domain filtering is performed using maximum pooling to obtain the filtered real and imaginary parts

$$\hat{X}'_{\text{real}} = f_{\text{MaxPool},s}\big(\hat{X}_{\text{real}}\big), \quad \hat{X}'_{\text{real}} \in \mathbb{R}^{C\times \frac{L}{2s}} \tag{1}$$

$$\hat{X}'_{\text{img}} = f_{\text{MaxPool},s}\big(\hat{X}_{\text{img}}\big), \quad \hat{X}'_{\text{img}} \in \mathbb{R}^{C\times \frac{L}{2s}} \tag{2}$$

where $s$ denotes the stride.

Then, the real and imaginary parts are concatenated and fed into the 1-D convolution, BN, and activation function

$$\hat{O} = \sigma\Big(f_{\text{conv,BN}}\Big(\Big[\hat{X}'_{\text{real}}, \hat{X}'_{\text{img}}\Big]\Big)\Big), \quad \hat{O} \in \mathbb{R}^{2C\times \frac{L}{2s}} \tag{3}$$

where $f_{\text{conv,BN}}(\bullet)$ denotes 1-D convolution and BN operation with kernel size $1 \times 1$ and stride 1, $\sigma(\bullet)$ is the LeakyReLU activation, and $[\bullet]$ is concatenation operation.

Then, $\hat{O}$ is split to obtain new real and imaginary parts, written as follows:

$$\hat{X}''_{\text{real}}, \ \hat{X}''_{\text{img}} = f_{\text{split}}\big(\hat{O}\big), \quad \hat{X}''_{\text{real}}, \ \hat{X}''_{\text{img}} \in \mathbb{R}^{C\times \frac{L}{2s}} \tag{4}$$

where $f_{\text{split}}(\bullet)$ denotes split operation.

Moreover, the new real part $\hat{X}''_{\text{real}}$ and imaginary part $\hat{X}''_{\text{img}}$ are transformed to the complex domain, and the final result is obtained by the inverse FFT (IFFT), written as follows:

$$O_{\text{Fourier}} = f_{\text{IFFT}}\Big(f_{\text{complex}}\Big(\hat{X}''_{\text{real}}, \hat{X}''_{\text{img}}\Big)\Big) \tag{5}$$

where $O_{\text{Fourier}} \in \mathbb{R}^{C\times(L/s)}$ is the final extracted feature of the global branch.

In the spatial domain of local branch, conventional 1-D convolution is used to capture information about the local details of the signal, which can be written as follows:

$$O_{\text{spatial}} = \sigma\big(f_{\text{conv,BN}}\big(\hat{X}\big)\big), \quad \hat{O} \in \mathbb{R}^{C\times \frac{L}{s}} \tag{6}$$

where $f_{\text{conv,BN}}(\bullet)$ denotes 1-D convolution and BN operation with kernel size $k \times 1$ and stride $s$ and $O_{\text{spatial}} \in \mathbb{R}^{C\times(L/s)}$ is the final extracted feature of the local branch.

Finally, the features of the two branches are summarized, and then, a $1 \times 1$ convolution is used to obtain the output features of the SFC block, which can be written as follows:

$$O_{\text{SFC}} = f_{\text{conv}}\big(O_{\text{spatial}} + O_{\text{Fourier}}\big) \tag{7}$$

where $O_{\text{SFC}}$ denotes the output features of the SFC block and $f_{\text{conv}}(\bullet)$ denotes convolution with kernel size $1 \times 1$.

### C. Two-Stage Training Strategy

*1) Adversarial Training:* The first stage of training uses adversarial learning to equip the GAN with initial fault detection capabilities. The generator then deceives the discriminator as much as possible. Assume that the training set $X = [X_{\text{normal}}, X_{\text{fault}}]$ consists of fault-free samples $X_{\text{normal}}$ and a small number of fault samples $X_{\text{fault}}$. The generator $G$ produces a sample that is treated as a fake sample by the discriminator $D$, while the real circuit signal $X$ is treated as a real sample by the discriminator $D$. Since the generator $G$ is a denoising autoencoder, we get $\hat{X} = [\hat{X}_{\text{normal}}, \hat{X}_{\text{fault}}]$ after adding noise $\eta$ to the training set. The corresponding sample produced by the generator can be denoted as $\widetilde{X} = [\widetilde{X}_{\text{normal}}, \widetilde{X}_{\text{fault}}]$. It should be noted that in the first training stage, it requires only fault-free samples to complete the training. The loss function for adversarial learning can be expressed as follows:

$$\min_G \max_D \big\{L_{G+D} = \mathbb{E}_{X_{\text{normal}}}\big[\log(1 - D(X_{\text{normal}}))\big] + \mathbb{E}_{\hat{X}_{\text{normal}}}\big[\log\big(D\big(G\big(\hat{X}_{\text{normal}}\big)\big)\big)\big]\big\} \tag{8}$$

where $E[\bullet]$ denotes expectation calculation. If we let 0 denote the real fault-free sample and 1 denote the fake sample, then the purpose of the generative network $G$ is to generate samples as close to true samples as possible, i.e., $D(G(\hat{X}_{\text{normal}}))$ is as close to 0 as possible. At this point, $L_{G+D}$ becomes smaller. The purpose of the discriminant network $D$ is to keep $D(X_{\text{normal}})$ close to 0 and $D(G(\hat{X}_{\text{normal}}))$ close to 1, at which point $L_{G+D}$ will increase.

In addition, the generator $G$ needs to reconstruct the noiseless samples, thus imposing an additional loss

$$L_G = \frac{1}{NM}\big\|X_{\text{normal}} - \tilde{X}_{\text{normal}}\big\|_2^2 \tag{9}$$

where $N$ represents the total number of fault-free samples, $M$ represents the length of samples, and $\| \bullet \|_2$ represents the calculation operation of $L_2$ distance.

Therefore, the loss function for generator $G$ can finally be expressed as follows:

$$L_{G_{\text{total}}} = L_G + \lambda L_{G+D} \tag{10}$$

where $\lambda$ is used to control the relative importance of the two terms.

*2) Supervised Learning:* In the second learning stage, the supervised learning paradigm is followed, and a limited number of labeled signals are used to train the proposed generator and discriminator to make them suitable for our target task. Moreover, the learning rate for training is halved, and a small number of epochs are used. The generator $G$ only guarantees the reconstruction effect of the fault-free samples and ignores the fault samples, so the loss function of the generator remains as (9). The purpose of this is to have the generator $G$ produce worse and more distinguishable reconstructions of faulty samples compared with normal samples, which helps the discriminator $D$ in identifying faults. The purpose of the discriminant network $D$ is to keep $D(G(\hat{X}_{\text{normal}}))$ close to 0 and $D(G(\hat{X}_{\text{fault}}))$ close to 1, which can be expressed as minimizing the following loss function:

$$L_D = \mathbb{E}_{\hat{X}_{\text{normal}}}\left[\log\left(D\left(G\left(\hat{X}_{\text{normal}}\right)\right)\right)\right] + \mathbb{E}_{\hat{X}_{\text{fault}}}\left[\log\left(1 - D\left(G\left(\hat{X}_{\text{fault}}\right)\right)\right)\right]. \quad (11)$$

### *D. IF Detection*

The detection threshold is determined by the discriminator output $D(G(\hat{X}_{\text{normal}}))$. Using a kernel density estimation (KDE) method, a threshold is obtained by taking a confidence limit of 0.95. More details about the KDE can be found in [31] and [32]. During the detection task, a fault is deemed to have occurred when the fault probability of the corresponding discriminator's output $D(G(\hat{X}_{\text{test}}))$ within the sliding window surpasses a predefined threshold value; otherwise, it is considered fault-free. Finally, IF detection results are obtained by testing each sliding window.

## III. Experimental Validation

### *A. Experimental Environment and Fault Setting*

Active filter circuits and power amplifier circuits are typical objects in the study of analog circuit IF detection. Active filter circuits have important role that they play in circuit protection, signal processing, and controller modules [1], [2], [3]. The main function of power amplifier circuits is to amplify low-power electrical signals to a sufficiently high power and is often used in communication systems, electric motors, radar systems, medical systems, and audio systems [40], [41]. Accordingly, the circuit under test (CUT) consists of two types, namely, the power amplifier circuit and the low-pass filter circuit. These two circuits (CUT-1 and CUT-2) are shown in Figs. 2 and 3. CUT-1 is set to a tolerance of 5% by Multisim. CUT-2 is obtained by testing in the actual platform shown in Fig. 4.

The input excitation of the CUT-1 is composed of 20-kHz, 0.5-V ac signal and 50-kHz, 0.5-V ac signal. The input excitation signal of the CUT-2 is composed of 5-kHz, 1-V ac signal and 10-kHz, 2-V ac signal. The acquisition signals for both circuits are selected from the voltage signals at the output points. The IF injection module consists of a sliding resistor and a voltage-controlled switch connected in parallel. It is connected in series to the corresponding position, and

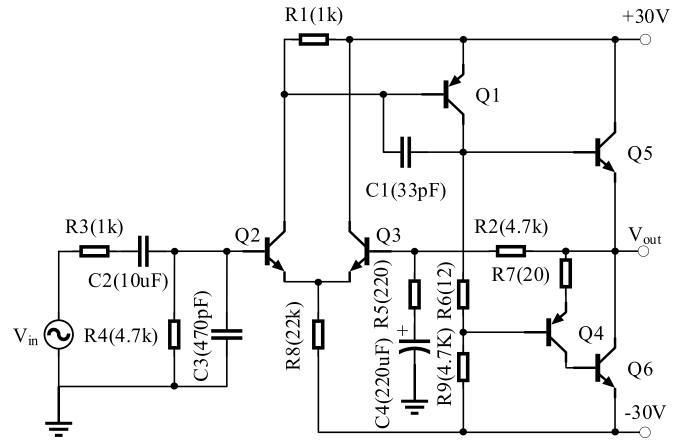


Fig. 2. CUT-1: power amplifier circuit.

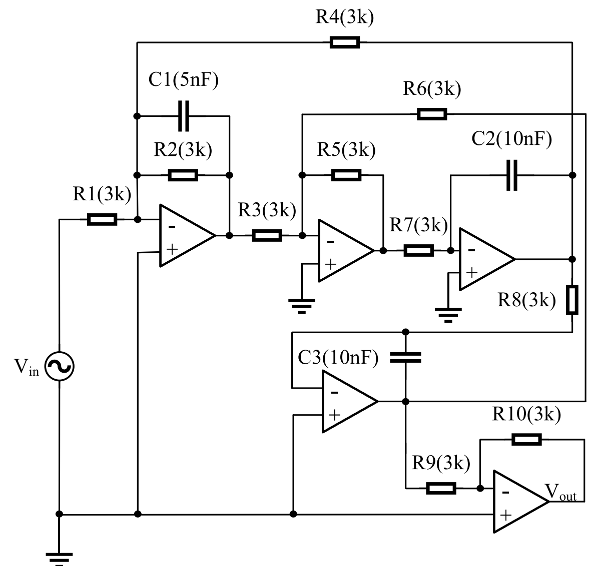


Fig. 3. CUT-2: low-pass filter.

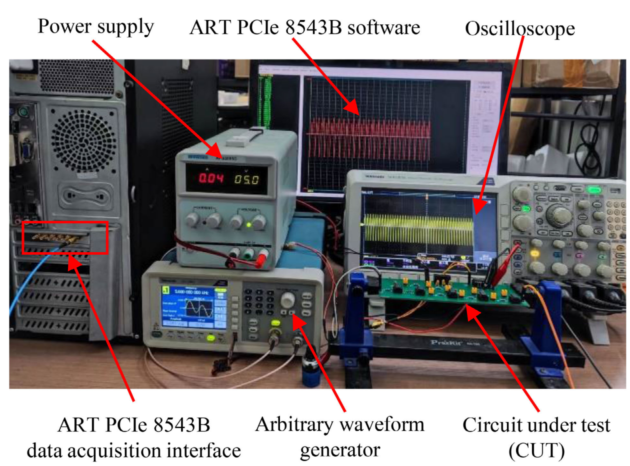


Fig. 4. Actual circuit experiment platform.

the duration of the IF injection is controlled by pulse width modulation (PWM) waves. Detailed fault settings are shown in Table I. In CUT-1 and CUT-2, nine and eight components, respectively, are considered to be key elements corresponding to the measurement points. In CUT-1, $Q1-e$ indicates the emitter of the triode. Since the faulty components in the same series line have the same effect on the output measurement points, there are two fuzzy sets in CUT-1, namely, $\{c2, r3\}$ and $\{c4, r5\}$. For millisecond IFs, each component is set to four fault levels based on different sizes of ECR.

### B. Model Training and Validation Strategy

A window length that is too large beyond the duration of the fault will result in an inability to accurately detect the start and end of the fault, and a window length that is too small will result in an increased computational burden. According to the description in Section II-A, the window length needs to be less than 1 ms. Given the sampling interval of 0.005 ms, this corresponds to a window length of fewer than 200 sampling points. In this study, a window length of 100 is selected, aligning with this criterion. Thus, the length of each sample is 100. The number of fault-free samples used for training in CUT-1 and CUT-2 is 7960 and 7998, respectively. The number of samples in the testing set is 95 760 and 80 000, respectively. In the first stage of training, the learning rates of the generator and discriminator are 0.0001 and 0.00005, respectively. The reason the generator's learning rate is higher than the discriminator's is to alleviate the problem of the generator's gradient vanishing due to the discriminator being trained too well [42]. The epoch and batch size are set to 50 and 512, respectively. The $\lambda$ in (10) is empirically taken as 1. In the second stage, which is a fine-tuning training, the learning rate of the generator and discriminator is halved compared with the previous. The epoch is 10. The batch size is set to the same size for each class of IFs. In order to test the performance of the model with different numbers of fault samples, numbers of 10, 30, and 50 are used in the second training stage for each type of fault. In addition, unless otherwise noted, the signal-to-noise ratio (SNR) is 10 dB to test the noise robustness of the different algorithms. All test results are averaged over five runs of the algorithm. The fault detection rate (FDR) and the false alarm rate (FAR) are utilized to evaluate the performance of IF detection, which are defined as follows:

$$\mathrm{FDR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{12}$$

$$\mathrm{FAR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{13}$$

where TP, FP, TN, and FN denote the respective numbers of true positive, false positive, true negative, and false negative samples. A superior detection performance is indicated by a higher FDR and a lower FAR.

### C. Parameter Analysis

Using the CUT-1 as an example, this section analyzes the parameters $\lambda$ in (10) and the epoch parameters of the two training stages in CUT-1. The detection performance curve combines FDR and FAR for all fault categories.

Fig. 5 shows the impact of the parameter $\lambda$ on the model's detection performance over the range $\{0.2, 0.4, \ldots, 5\}$. It can be seen that the FAR remains essentially at 5%, and the FDR stays at a high level in the range $[1, 2)$, and the FDR is lowest in the interval $[0.2, 1)$. After $\lambda > 2$, the FDR shows a slight decline but stabilizes for the most part. Overall, the performance does not show significant fluctuations.

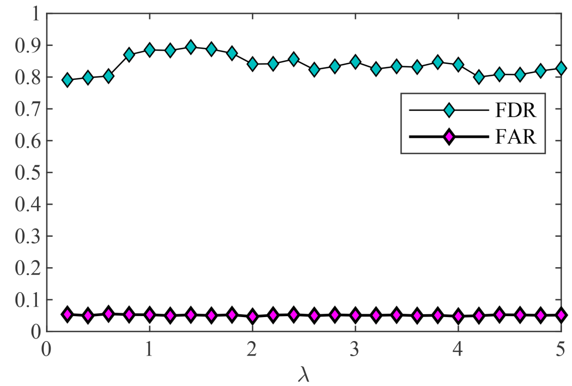


Fig. 5. Detection performance with different values of $\lambda$.

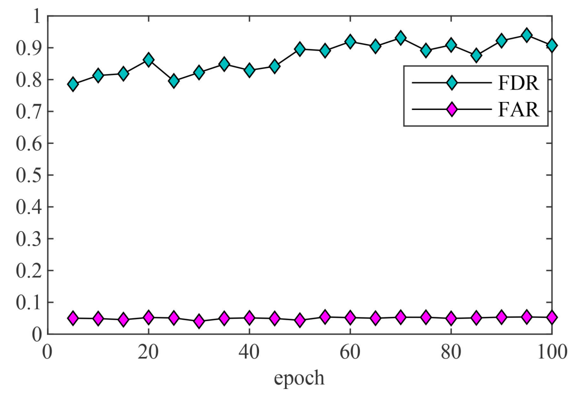


Fig. 6. Detection performance at different epochs in training stage 1.

Fig. 6 shows the impact of epoch in the range $\{5, 10, \ldots, 100\}$ on the model's detection performance during training stage 1. In the experimental process, the number of epochs for training the second stage is fixed, while only the number of epochs for the first stage is changed. It can be observed that FDR shows a slight increasing trend, and after reaching 50 epochs, FDR remains at around 90%, while FAR stabilizes at about 5%.

Fig. 7 shows the impact of epoch in the range $\{5, 10, \ldots, 100\}$ on the model's detection performance during training stage 2. During the experiment, the number of epochs for training the first stage is fixed, with only the number of epochs for the second stage being altered. It can be seen that, due to the pretraining in the first stage, FDR achieves a high level with very few epochs, and after epoch 10, FDR essentially stabilizes. Therefore, only a small number of epochs are needed for the second stage.

### D. Effectiveness of the Proposed Method

*1) Influence Under Different Numbers of Fault Samples:* To investigate the impact of using a limited number of fault samples during training on the performance of the proposed

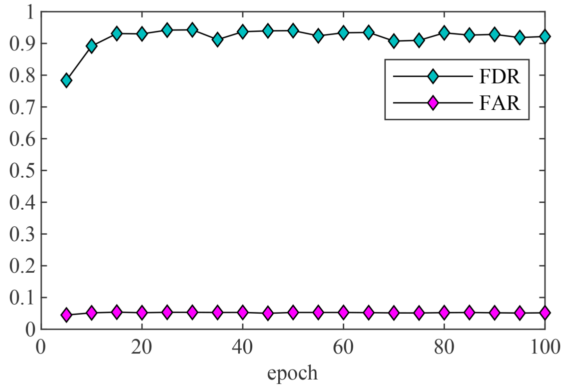


Fig. 7. Detection performance at different epochs in training stage 2.

TABLE I
FAULT SETTINGS

| Circuit | Fault component | Fault level: ECR | Duration |
|---|---|---|---|
| CUT-1 | {r2}, {c2,r3}, {c4,r5}, {r6}, {Q1-e}, {Q2-e}, {Q3-e} | I: 0.5 kΩ<br>II: 1 kΩ<br>III: 1.5 kΩ<br>IV: 2 kΩ | 1-5 ms |
| CUT-2 | {r4}, {r6}, {r7}, {r8}, {r9}, {r10}, {c2}, {c3} | | |

TABLE II
FDRS (%) OF THE PROPOSED METHOD WITH DIFFERENT SAMPLE NUMBERS FOR EACH FAULT TYPE IN CUT-1

| Sample size | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 | $76.16 \pm 0.78$ | $87.11 \pm 1.52$ | $96.27 \pm 1.25$ | $99.35 \pm 0.46$ |
| 30 | $81.30 \pm 0.74$ | $95.31 \pm 0.78$ | $99.41 \pm 0.17$ | $99.95 \pm 0.03$ |
| 50 | $84.22 \pm 0.61$ | $97.52 \pm 0.27$ | $99.83 \pm 0.04$ | $99.98 \pm 0.01$ |

TABLE III
FDRS (%) OF THE PROPOSED METHOD WITH DIFFERENT SAMPLE NUMBERS FOR EACH FAULT TYPE IN CUT-2

| Sample size | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 | $64.36 \pm 2.14$ | $86.48 \pm 3.07$ | $93.83 \pm 1.57$ | $96.23 \pm 0.42$ |
| 30 | $72.22 \pm 2.71$ | $92.45 \pm 1.39$ | $96.08 \pm 0.41$ | $96.78 \pm 0.07$ |
| 50 | $76.66 \pm 1.15$ | $93.98 \pm 1.09$ | $96.38 \pm 0.31$ | $96.87 \pm 0.05$ |

model, only 10, 30, and 50 samples for each class of faults are used for training, and the test results are obtained accordingly. The test results are shown in Tables II–V. As can be seen from Tables IV and V, the FAR of our proposed method is around 5% in both CUT-1 and CUT-2. In Tables II and III, the FDR of the proposed method is enhanced, as the number of fault samples used for training increases. Moreover, while maintaining a small fluctuation of the FAR, the FDR increases, as the fault level is enhanced. This indicates that as the ECR increases, the output deviation to the circuit begins to increase, resulting in IFs that are easier to detect than before. Fig. 8 shows the detection curve for $r3$ in CUT-1 at different fault levels. The blue detection curve is above the red threshold line representing the detection of a fault. It can be noticed that as the fault level increases, the detection curve gets closer to 1 at the moment of the fault. Fig. 9 shows the detection curve for $c3$ in CUT-2 for different numbers of training samples. Observably, with an increase in training samples, the fluctuation of the detection curve during fault-free periods diminishes, and the amplitude of the detection curve during fault occurrences rises.

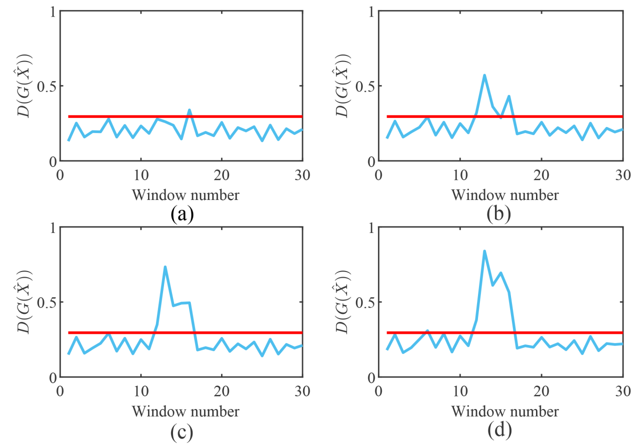


Fig. 8. Detection curve for $r3$ in CUT-1 at different fault levels. (a) Fault level I. (b) Fault level II. (c) Fault level III. (d) Fault level IV.

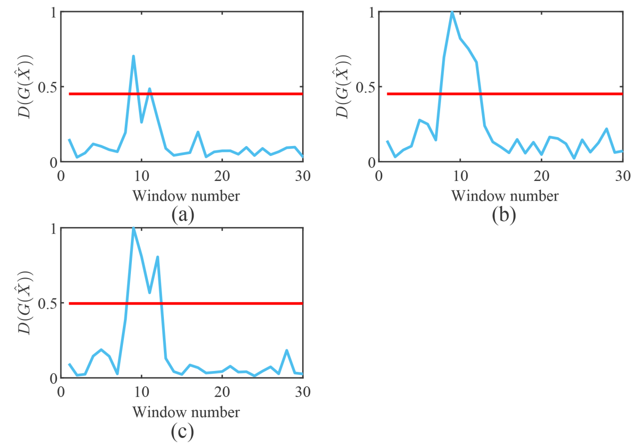


Fig. 9. Detection curve for $c3$ in CUT-2 under different numbers of training samples. (a) Ten samples per category. (b) Thirty samples per category. (c) Fifty samples per category.

TABLE IV
FARS (%) WITH DIFFERENT SAMPLE NUMBERS FOR EACH FAULT TYPE IN CUT-1

| Sample size | Ours | Network-1 | Network-2 | Network-3 |
|---|---|---|---|---|
| 0 | / | / | $5.58 \pm 0.59$ | / |
| 10 | $5.15 \pm 0.29$ | $5.22 \pm 0.20$ | / | $10.13 \pm 1.31$ |
| 30 | $5.14 \pm 0.09$ | $5.28 \pm 0.17$ | / | $6.67 \pm 1.43$ |
| 50 | $6.06 \pm 0.79$ | $5.30 \pm 0.19$ | / | $8.04 \pm 1.86$ |

*2) Influence Under Different SNRs:* In order to test the effect of different noise intensities on the performance of the proposed model, the two circuits is tested at three SNRs (10, 15, and 20 dB). For training, the number of fault samples is fixed at 10 for each class. The test results are shown in Tables VI–VIII. It can be seen that the FDR gradually increases, as the SNR increases. For example, at fault level I of 15 dB, the FDR of CUT-2 is improved by 19.68% compared with 10 dB, with little difference in the FAR. In two circuits,

TABLE V
FARs (%) WITH DIFFERENT SAMPLE NUMBERS FOR EACH FAULT TYPE IN CUT-2

| Sample size | Ours | Network-1 | Network-2 | Network-3 |
|---|---|---|---|---|
| 0 | / | / | $7.64 \pm 1.53$ | / |
| 10 | $5.33 \pm 0.26$ | $3.38 \pm 0.29$ | / | $5.13 \pm 0.48$ |
| 30 | $5.90 \pm 0.17$ | $3.73 \pm 0.31$ | / | $5.71 \pm 0.34$ |
| 50 | $6.02 \pm 0.23$ | $4.29 \pm 0.18$ | / | $5.71 \pm 0.23$ |

TABLE VI
FDRs (%) OF THE PROPOSED METHOD WITH DIFFERENT SNRs IN CUT-1

| SNR | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 dB | $76.16 \pm 0.78$ | $87.11 \pm 1.52$ | $96.27 \pm 1.25$ | $99.35 \pm 0.46$ |
| 15 dB | $80.17 \pm 1.93$ | $94.13 \pm 2.76$ | $99.11 \pm 0.67$ | $99.85 \pm 0.24$ |
| 20 dB | $83.71 \pm 0.99$ | $97.63 \pm 0.68$ | $99.91 \pm 0.08$ | $100 \pm 0$ |

TABLE VII
FDRs (%) OF THE PROPOSED METHOD WITH DIFFERENT SNRs IN CUT-2

| SNR | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 dB | $64.36 \pm 2.14$ | $86.48 \pm 3.07$ | $93.83 \pm 1.57$ | $96.23 \pm 0.42$ |
| 15 dB | $84.04 \pm 1.81$ | $96.02 \pm 0.54$ | $96.85 \pm 0.09$ | $96.96 \pm 0.03$ |
| 20 dB | $93.82 \pm 0.57$ | $96.97 \pm 0.03$ | $96.95 \pm 0.04$ | $97.06 \pm 0.04$ |

TABLE VIII
FARs (%) OF THE PROPOSED METHOD WITH DIFFERENT SNRs

| Circuit | 10 dB | 15 dB | 20 dB |
|---|---|---|---|
| CUT-1 | $5.15 \pm 0.29$ | $5.67 \pm 0.05$ | $5.58 \pm 0.21$ |
| CUT-2 | $5.33 \pm 0.26$ | $5.64 \pm 0.12$ | $5.39 \pm 0.41$ |

TABLE IX
FDRs (%) OF THE PROPOSED METHOD WITHOUT TRAINING STAGE 1 IN CUT-1

| Sample size | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 | $73.58 \pm 1.30$ | $78.92 \pm 2.17$ | $86.06 \pm 3.57$ | $91.77 \pm 4.45$ |
| 30 | $77.67 \pm 1.94$ | $90.39 \pm 4.18$ | $97.59 \pm 2.76$ | $99.28 \pm 1.09$ |
| 50 | $80.37 \pm 0.88$ | $94.76 \pm 1.00$ | $99.26 \pm 0.39$ | $99.91 \pm 0.07$ |

TABLE X
FDRs (%) OF THE PROPOSED METHOD WITHOUT TRAINING STAGE 1 IN CUT-2

| Sample size | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 | $24.03 \pm 4.20$ | $41.09 \pm 6.29$ | $52.16 \pm 6.94$ | $56.80 \pm 7.51$ |
| 30 | $37.14 \pm 6.91$ | $56.70 \pm 8.74$ | $68.89 \pm 7.64$ | $74.84 \pm 6.56$ |
| 50 | $54.41 \pm 3.74$ | $74.05 \pm 2.95$ | $82.17 \pm 3.52$ | $86.23 \pm 4.10$ |

the FDR is less than 80% at fault level I of 10 dB, indicating that the SNR of 10 dB and fault level I are the performance limits of the model. However, the proposed model remains ahead of the state-of-the-art methods in the subsequent comparative validation.

*3) Effectiveness of Training Stage 1 for Adversarial Learning:* In order to verify the effectiveness of the training stage 1, Network-1 is constructed in this section. Compared with the proposed method, Network-1 removes the training strategy of adversarial learning and retains only the second stage of supervised learning. Tables IV, V, IX, and X demonstrate the performance of Network-1. The FAR for Network-1 in CUT-1 is not much different from the proposed method, and there is a small decrease in the FAR in CUT-2. However, compared with the FDRs of the proposed methods in Tables II and III, Network-1 shows a significant decrease in the FDR. This phenomenon is even more pronounced in CUT-2. For example, the FDR of fault level I in CUT-2 under ten fault samples decreased by 40.33%. This indicates that adversarial learning in training stage 1 can provide the model with initial detection capabilities and helps to reduce the dependence on the number of fault samples for training.

TABLE XI
FDRs (%) OF THE PROPOSED METHOD WITHOUT TRAINING STAGE 2

| Circuit | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| CUT-1 | $66.90 \pm 4.96$ | $75.89 \pm 6.73$ | $83.72 \pm 7.71$ | $90.26 \pm 6.80$ |
| CUT-2 | $24.77 \pm 4.74$ | $53.05 \pm 9.88$ | $68.39 \pm 10.97$ | $80.09 \pm 10.40$ |

TABLE XII
FDRs (%) OF THE PROPOSED METHOD WITHOUT THE SFC BLOCK IN CUT-1

| Sample size | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 | $71.28 \pm 7.10$ | $79.74 \pm 3.59$ | $82.97 \pm 2.79$ | $83.60 \pm 2.62$ |
| 30 | $76.30 \pm 0.73$ | $81.08 \pm 1.72$ | $85.21 \pm 2.33$ | $86.82 \pm 2.81$ |
| 50 | $77.88 \pm 1.47$ | $83.37 \pm 1.90$ | $86.54 \pm 2.55$ | $87.35 \pm 3.40$ |

TABLE XIII
FDRs (%) OF THE PROPOSED METHOD WITHOUT THE SFC BLOCK IN CUT-2

| Sample size | Fault level I | Fault level II | Fault level III | Fault level IV |
|---|---|---|---|---|
| 10 | $64.16 \pm 2.17$ | $79.34 \pm 2.38$ | $87.41 \pm 2.75$ | $92.59 \pm 2.49$ |
| 30 | $70.77 \pm 1.75$ | $83.00 \pm 4.06$ | $87.99 \pm 4.72$ | $90.76 \pm 4.44$ |
| 50 | $71.15 \pm 0.79$ | $83.71 \pm 2.88$ | $89.75 \pm 3.23$ | $93.55 \pm 2.23$ |

*4) Effectiveness of Training Stage 2 for Supervised Learning:* Network-2 is used to validate the training stage 2. Network-2 removes the second stage of training in the proposed method, which is equivalent to training without using any fault samples. As can be seen in Tables IV, V, and XI, there is a significant degradation in Network-2 performance. This indicates that supervised learning with fewer samples in training stage 2 is beneficial in guiding the model to detect IFs and discover fault information hidden in the noise.

*5) Effectiveness of the SFC Block:* To verify the effectiveness of the SFC block, Network-3 replaces the SFC block with a 1-D convolution module. The detection results of Network-3 are shown in Tables IV, V, XII, and XIII. Without the help of the SFC block, the FARs in CUT-1 have increased, and the robustness to noise has weakened. The FDRs for both circuits also show some decrease compared with those in Tables II and III. This demonstrates that learning local and global features from the spatial and Fourier domains facilitates learning local details of the signal under noise and expanding the receptive field.

TABLE XIV
FDRs (%) With Different State-of-the-Art Deep Learning Methods

| Circuit | Fault level | Ours | SSRL | PKGTS | OIG | GBAD | AMDSCAE | DVAE |
|---|---|---|---|---|---|---|---|---|
| CUT-1 | I | $\mathbf{76.16 \pm 0.78}$ | $74.35 \pm 1.28$ | $74.22 \pm 0.35$ | $53.89 \pm 4.52$ | $74.53 \pm 0.69$ | $68.71 \pm 8.32$ | $72.43 \pm 0.96$ |
| | II | $\mathbf{87.11 \pm 1.52}$ | $82.40 \pm 5.23$ | $80.51 \pm 2.05$ | $72.39 \pm 5.30$ | $83.93 \pm 1.27$ | $85.83 \pm 5.74$ | $75.87 \pm 0.93$ |
| | III | $\mathbf{96.27 \pm 1.25}$ | $88.66 \pm 7.40$ | $88.56 \pm 2.70$ | $79.39 \pm 4.24$ | $92.19 \pm 1.36$ | $93.27 \pm 4.13$ | $81.21 \pm 1.63$ |
| | IV | $\mathbf{99.35 \pm 0.46}$ | $90.54 \pm 7.34$ | $94.06 \pm 3.02$ | $81.74 \pm 3.06$ | $96.71 \pm 0.88$ | $97.15 \pm 2.11$ | $88.60 \pm 1.63$ |
| CUT-2 | I | $\mathbf{64.36 \pm 2.14}$ | $64.12 \pm 6.56$ | $35.43 \pm 6.65$ | $41.68 \pm 8.02$ | $35.58 \pm 2.34$ | $35.13 \pm 3.46$ | $27.59 \pm 5.57$ |
| | II | $\mathbf{86.48 \pm 3.07}$ | $79.26 \pm 3.23$ | $68.21 \pm 9.64$ | $64.86 \pm 7.38$ | $67.98 \pm 2.93$ | $69.66 \pm 4.53$ | $58.60 \pm 8.74$ |
| | III | $\mathbf{93.83 \pm 1.57}$ | $83.46 \pm 2.80$ | $83.40 \pm 8.45$ | $76.36 \pm 6.68$ | $82.57 \pm 1.91$ | $84.72 \pm 2.83$ | $73.23 \pm 7.18$ |
| | IV | $\mathbf{96.23 \pm 0.42}$ | $86.09 \pm 3.68$ | $90.59 \pm 6.85$ | $83.88 \pm 6.08$ | $90.09 \pm 0.92$ | $91.86 \pm 1.58$ | $80.71 \pm 4.66$ |

TABLE XV
FARs (%) With Different State-of-the-Art Deep Learning Methods

| Circuit | Ours | SSRL | PKGTS | OIG | GBAD | AMDSCAE | DVAE |
|---|---|---|---|---|---|---|---|
| CUT-1 | $5.15 \pm 0.29$ | $6.51 \pm 0.67$ | $5.82 \pm 0.2$ | $7.23 \pm 1.36$ | $5.61 \pm 0.34$ | $5.04 \pm 0.72$ | $5.31 \pm 0.09$ |
| CUT-2 | $5.33 \pm 0.26$ | $5.48 \pm 0.67$ | $5.87 \pm 0.35$ | $6.82 \pm 0.93$ | $6.04 \pm 0.59$ | $5.59 \pm 0.35$ | $5.47 \pm 0.67$ |

TABLE XVI
Training and Testing Time of Networks

| | Ours | SSRL | PKGTS | OIG | GBAD | AMDSCAE | DVAE |
|---|---|---|---|---|---|---|---|
| Training Time/s | 78.277 | 31.063 | 148.437 | 36.313 | 61.251 | 80.996 | 26.332 |
| Testing Time/ms | 0.080 | 0.093 | 0.088 | 0.073 | 0.087 | 0.097 | 0.094 |

### E. Compared With State-of-the-Art Deep Learning Methods

In this section, six state-of-the-art methods of detection are used to compare the proposed methods, including SSRL [33], PKGTS [32], OIG [36], GBAD [35], AMDSCAE [31], and DVAE [30]. For the problem of limited fault data, these six approaches have made efforts from different perspectives. During the comparison, both our proposed method and the SSRL method used only ten samples per fault class for training. Moreover, the OIG method needs to replace the 2-D convolution with a 1-D convolution to accommodate the signal patterns in this article. Noise is added during testing to make the SNR of 10 dB.

The comparison results are shown in Tables XIV and XV. It can be found that the FARs of the OIG method are slightly higher, and the FARs of the other methods basically remain around 5%. In addition, the FDRs of the proposed method has a significant advantage. For example, compared with the second effective SSRL method in CUT-2, the proposed method exhibits improvement in FDRs by 0.24%, 7.22%, 10.37%, and 10.14% for different fault levels, respectively. This shows that the proposed method has better results under the condition of limited fault samples and facilitates the detection of IFs.

### F. Computational Burden of the Networks

Since IFs have a short duration, it is necessary for the algorithm to be highly efficient during online testing. Training, on the other hand, can be done offline, so the computational efficiency requirements for the training stage are not as high as for the testing stage. Using CUT-1 as an example, the training time represents the cumulative time for all samples, while the testing time denotes the average time spent detecting each sample. Table XVI displays the training and testing times of the proposed method and the comparative methods. Due to the need to train two models in PKGTS, it took the longest training time, and the multibranch structure of AMDSCAE increased the complexity, taking 80.996 s for training. The proposed method requires training in two stages, and as a result, the training time is relatively long, totaling 78.277 s. During the testing stage, the proposed method has a detection time second only to the OIG method, with an average time of 0.080 ms per sample, which is sufficient for millisecond-level fault detection. Compared with the time-consuming AMDSCAE, the detection time is reduced by 17.53%. The proposed method leads all compared methods in detection accuracy while maintaining detection speed.

## IV. Conclusion

This article investigates the utilization of GAN in the realm of IF detection for analog circuits. The proposed two-stage training strategy and SFC block can effectively improve fault detection. The two-stage training strategy initially prepares the model for detection without using fault data via adversarial learning. Subsequently, a small amount of fault data is employed to guide the model's perception of faults through supervised learning. The proposed SFC block learns the local details and global information of the signal through the spatial and Fourier domains, respectively, thus enhancing the detection performance and noise robustness. The effectiveness of the proposed method is demonstrated through studies based on two typical circuits. In future work, we will focus on further optimizing the algorithm's operational efficiency to meet the demands of real-time detection scenarios.

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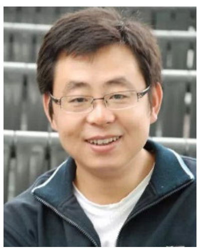

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